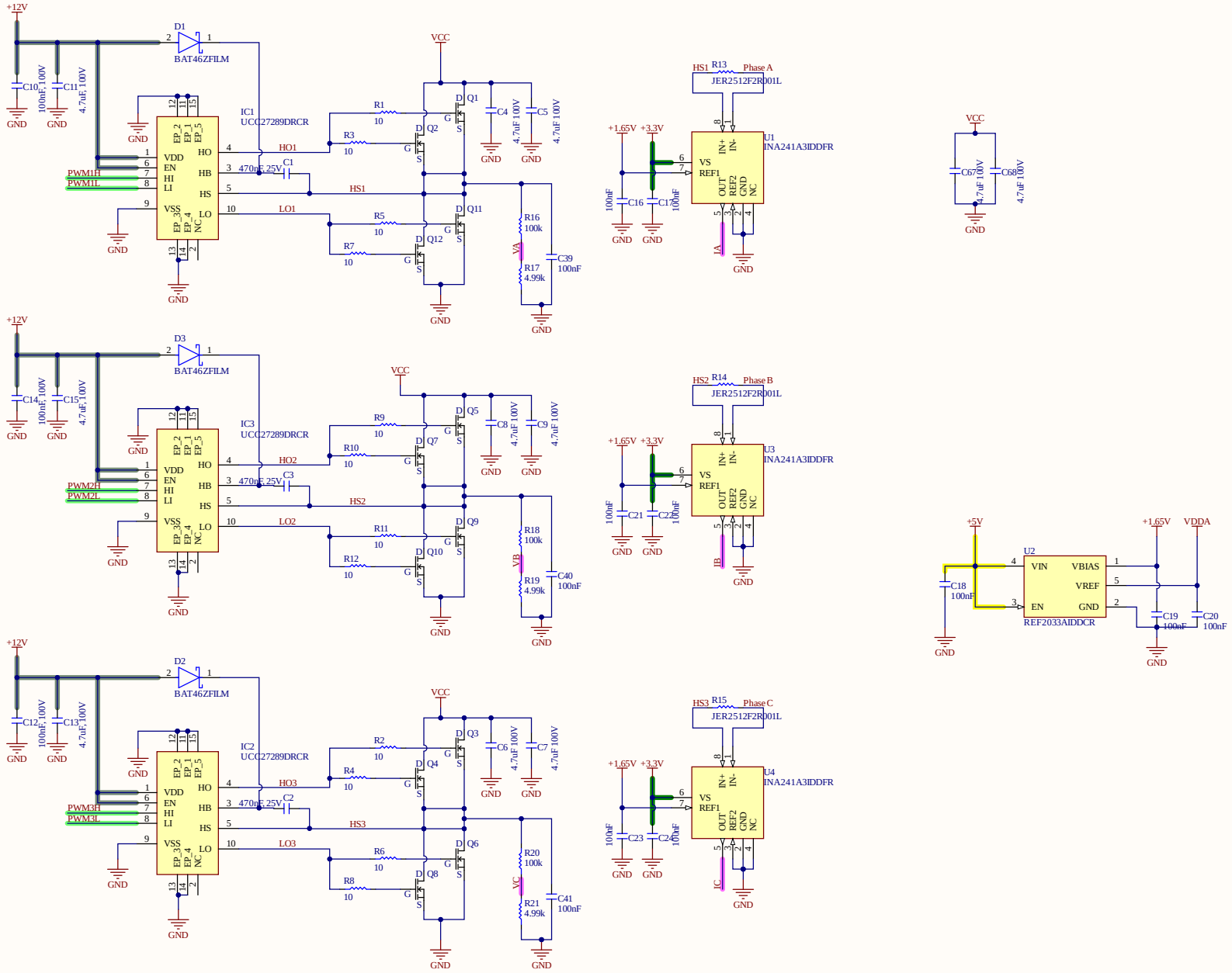
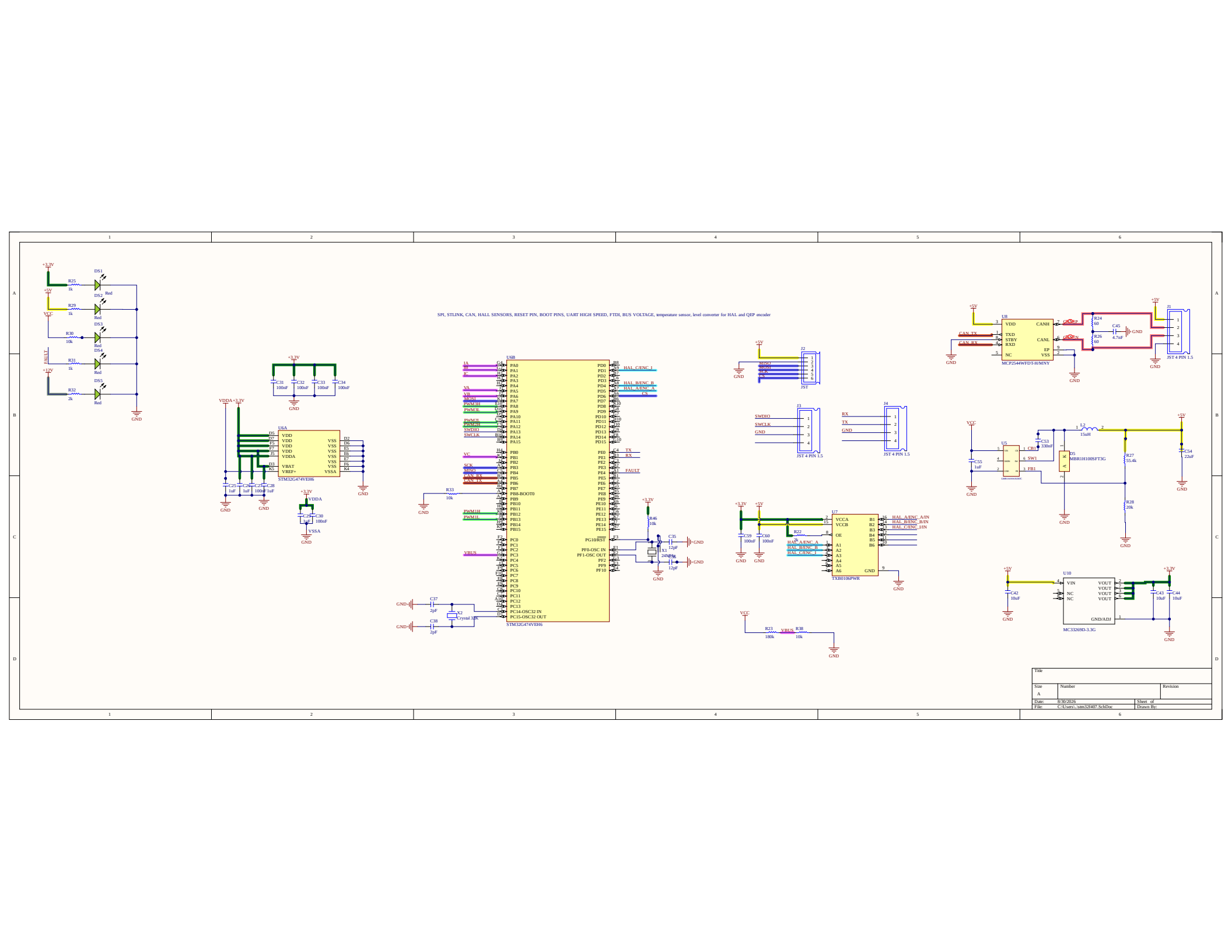
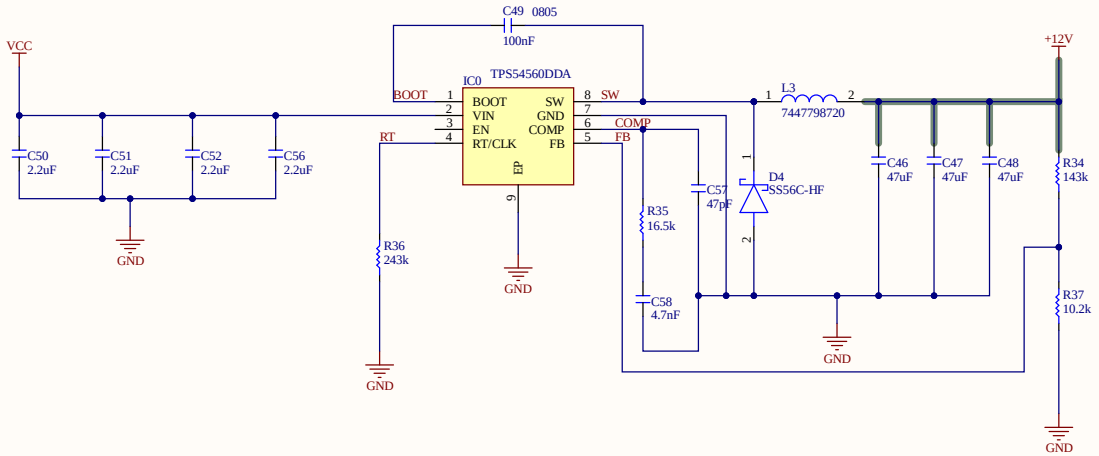
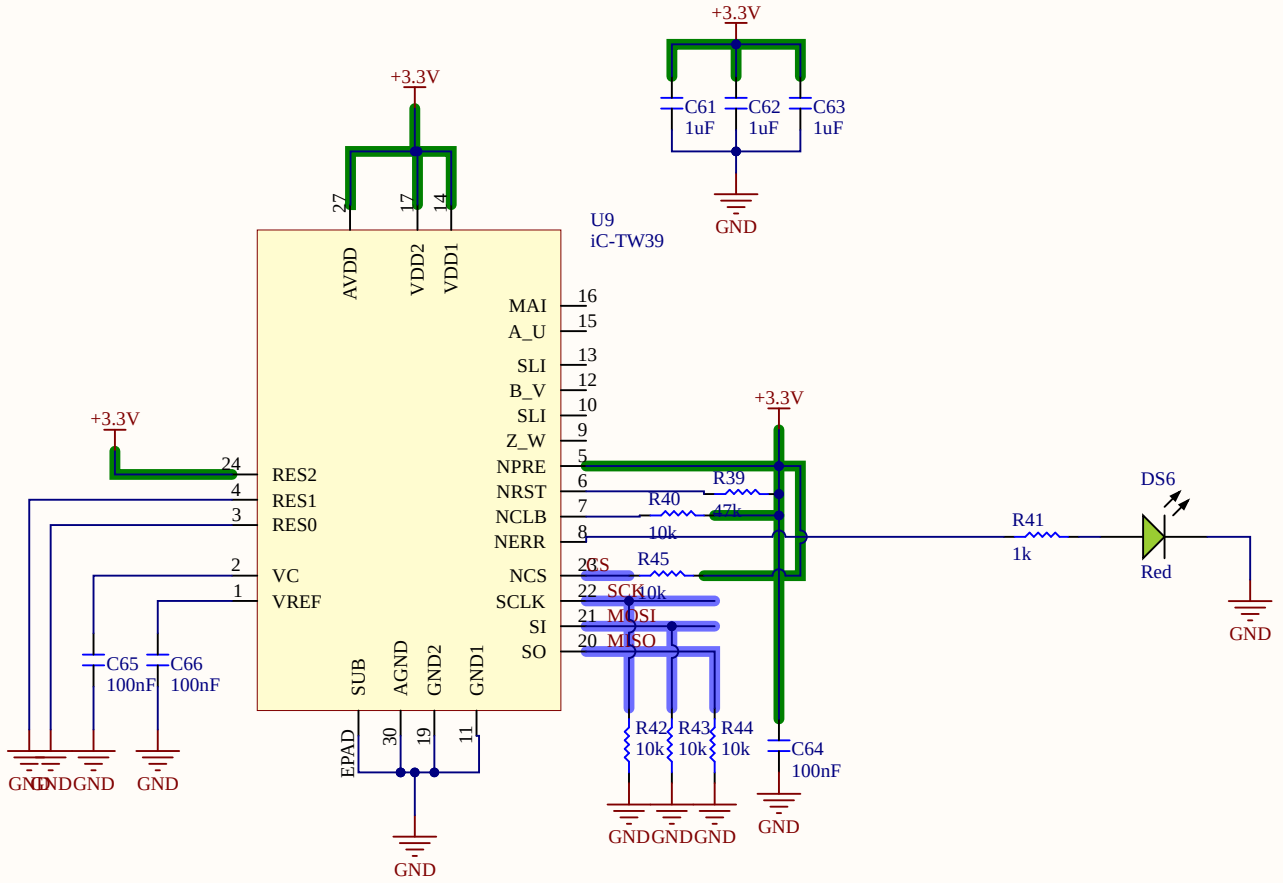


Title		
Size: A	Number	Revision
Date: 8/30/2026		Sheet of
File: C:\Users\..\Inverter.SchDoc		Drawn By:


SPI, STLINK, CAN, HALL SENSORS, RESET PIN, BOOT PINS, UART HIGH SPEED, FTDI, BUS VOLTAGE, temperature sensor, level converter for HAL and QEP encoder
STM32G474VEH6
MCP2544WFDT-H/MNY
TXB0106PWR
MC33269D-3.3G
JST 4 PIN 1.5
Title
Size
Number
Revision
Date: 8/30/2026
File: C:\Users\..\stm32f407.SchDoc
Sheet of
Drawn By:



Title	Buck Converter		
Size: B	Number:	Revision: V1	
Date: 8/30/2026		Sheet of	
File: C:\Users\..\Buck Converters.SchDoc		Drawn By: S Imtiaz Ali Shah	



Title		
Size: A	Number	Revision
Date: 8/30/2026		Sheet of
File: C:\Users\..\Encoder.SchDoc		Drawn By:

Name	Description	Designator	Footprint	LibRef	Quantity
470nF, 25V		C1, C2, C3	C_0805	C_0805	3
4.7uF 100V		C4, C5, C6, C7, C8, C9, C67, C68	C_1210	C_1210	8
100nF, 100V		C10, C12, C14	C_0805	C_0805	3
4.7uF, 100V		C11, C13, C15	C_0805	C_0805	3
100nF		C16, C17, C18, C19, C20, C21, C22, C23, C24, C39, C40, C41, C49	C_0603	C_0603	13
1uF		C25, C26, C28, C29, C55, C61, C62, C63	C_0402	C_0402	8
100nF		C27, C30, C31, C32, C33, C34, C59, C60, C64, C65, C66	C_0402	C_0402	11
12pF		C35, C36	C_0402	C_0402	2
2pF		C37, C38	C_0402	C_0402	2
10uF		C42, C43, C44	C_0805	C_0805	3
4.7nF		C45, C58	C_0402	C_0402	2
47uF		C46, C47, C48	C_0603	C_0603	3
2.2uF		C50, C51, C52, C56	C_0603	C_0603	4
330nF		C53	C_0402	C_0402	1
22uF		C54	C_0402	C_0402	1
47pF		C57	C_0402	C_0402	1
BAT46ZFILM	DIODE SCHOTTKY 100V 150MA SOD123	D1, D2, D3	FP-SOD-123-MFG	CMP-12187-000006-1	3
SS56C-HF	Schottky Diode	D4	DIOM7859X239N	SS56C-HF	1
MBR1H100SFT3G	Schottky Power Rectifier, 2-Pin SOD-123, Pb-Free, Tape and Reel	D5	ONSC-SOD-123FL-2-498-01_V	CMP-1055-00439-1	1
Red	Chip LED, Yellow-Green, 5 V, 25 mA, -40 to 85 degC, 2-Pin SMD (0603), RoHS, Tape and Reel	DS1, DS2, DS3, DS4, DS5, DS6	EVRL-0603-2_V	CMP-007-00002-2	6
TPS54560DDA	Integrated Circuit	IC0	SOIC127P600X170-9N	TPS54560DDA	1
UCC27289DRCR	Integrated Circuit	IC1, IC2, IC3	TPS51200MDRCTEP	UCC27289DRCR	3
JST 4 PIN 1.5		J1, J3, J4	JST 4 PIN 1.5	JST 4 PIN 1.5	3
JST		J2	JST 6 PIN 1.5	JST 6 PIN 1.5	1
15uH	Fixed Inductors Commercial High Saturation Series 15uH 1.9A 265mΩ 20%	L2	FP-IHLP-2020CZ-FP_1_905x2-MFG	CMP-02424-000144-1	1
7447798720	Inductor	L3	7447797360	7447798720	1
BSZ075N08NS5ATMA1	MOSFET (N-Channel)	Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8, Q9, Q10, Q11, Q12	BSZ0909LSATMA1	BSZ075N08NS5ATMA1	12
10		R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12	R_0402	R_0402	12
JER2512F2R001L	RES 1mΩ ±1% 2W 2512	R13, R14, R15	JER2512F2R001L	JER2512F2R001L	3
100k		R16, R18, R20	R_0402	R_0402	3
4.99k		R17, R19, R21	R_0402	R_0402	3
2k		R22, R32	R_0402	R_0402	2
180k		R23	R_0603	R_0603	1
60		R24, R26	R_0402	R_0402	2
1k		R25, R29, R31, R41	R_0402	R_0402	4
55.4k		R27	R_0402	R_0402	1
20k		R28	R_0402	R_0402	1
10k		R30, R33, R40, R42, R43, R44, R45, R46	R_0402	R_0402	8
143k		R34	R_0402	R_0402	1
16.5k		R35	R_0402	R_0402	1
243k		R36	R_0402	R_0402	1
10.2k		R37	R_0402	R_0402	1
10k		R38	R_0603	R_0603	1
47k		R39	R_0402	R_0402	1
INA241A3IDDFR	-5-V to 110-V bidirectional ultraprecise current sense amplifier with enhanced PWM rejection	U1, U3, U4	SOT65P280X110-8N	INA241A3IDDFR	3
REF2033AIDDCR	IC VREF SERIES 1.65V/3.3V 5SOT	U2	FP-DDC0005A-MFG	CMP-04916-000008-1	1
LMR16006XDDC		U5	SOT95P280X110-6N	LMR16006XDDC	1
STM32G474VEH6	IC MCU 32BIT 512KB FLSH 100TFBGA	U6	FP-TFBGA100-A08Q-MFG	CMP-12105-000111-1	1
TXB0106PWR		U7	Texas_Instruments_PW0016A_MFG	CMP-2000-07336-2	1
MCP2544WFDT-H/MNY	IC TRANSCEIVER 1/1 8TDFN	U8	FP-C04-129-MNY-MFG	CMP-06279-000053-1	1
iC-TW39		U9	iC-TW39	iC-TW39	1
MC33269D-3.3G		U10	* SOIC127P600X175-8N	MC33269D-3.3G	1
24MHz		X1	24MHz Crystal	24MHz Crystal	1
Crystal 32K		X2	Crystal 32k	Crystal 32K	1